usb_power
USB−C sink (no host), VBUS protection, USB0 device/FEL
File: usb_power.kicad_sch

power
Regulators, rail order, supervisors, staged−power links
File: power.kicad_sch

f133
F133−B SoC: supplies, clock, reset, pin assignment
File: f133.kicad_sch

boot_nor
32 MiB SPI NOR boot (BOOT−SEL 00), straps, FEL/clip recovery
File: boot_nor.kicad_sch

mn820
MN820 (BT820B + 128 MiB DDR3L) module socket, host SPI1, LVDS to R09 panel
File: mn820.kicad_sch

fpga
LFE5U−45F−6BG381C (optional accelerator): rails, sysCONFIG slave SPI, JTAG, SPI0 isolation
File: fpga_ecp5.kicad_sch

psram
APS6404L 8 MiB PSRAM: switched VDD (DONE−gated), static pulls
File: psram.kicad_sch

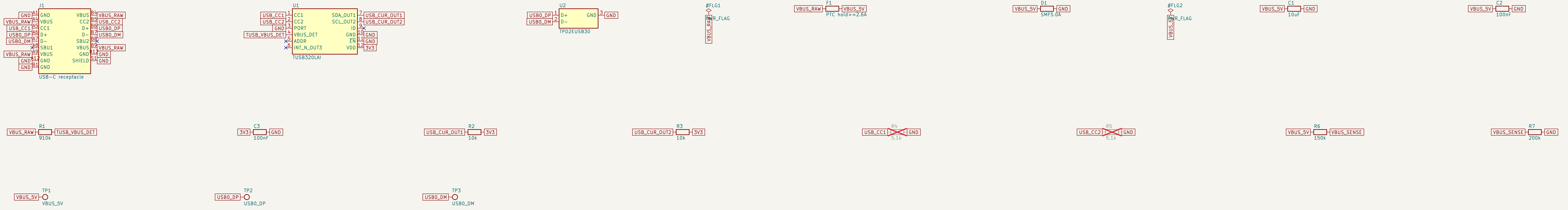
sd_audio
microSD game media (not boot), 2x MAX98357A I2S speakers
File: sd_audio.kicad_sch

controls
Game buttons, LEDs, debug UART
File: controls.kicad_sch

NOT approved for fabrication. LCD/panel/backlight deferred to R09.

Sheet: /
File: ARC1_R85.kicad_sch
Title: ARC−1 Compact R8.5 − hierarchical root
Size: A3
Date: 2026−10−15
Rev: R85−draft
KiCad E.D.A. 9.0.9
Id: 1/10

USB-C sink (no host), VBUS protection, USB0 device/FEL
Sheet generated from design.py; global labels carry every net.


J1
USB-C receptacle
GND
VBUS
CC1
D+
D−
SBU1
VBUS
GND
GND
VBUS
CC2
D+
D−
SBU2
VBUS
GND
SHIELD
A1
A4
A5
A6
A7
A8
A9
A12
B1
B4
B5
B6
B7
B8
B9
B12
S1
VBUS_RAW
USB_CC1
USB_CC2
USB0_DP
USB0_DM
GND
U1
TUSB320LAI
CC1
CC2
PORT
VBUS_DET
ADDR
INT_N_OUT3
SDA_OUT1
SCL_OUT2
ID
GND
EN
VDD
TUSB_VBUS_DET
USB_CUR_OUT1
USB_CUR_OUT2
3V3
U2
TPD2EUSB30
D+
D−
GND
#FLG1
PWR_FLAG
F1
PTC hold>=2.6A
VBUS_5V
D1
SMF5.0A
#FLG2
PWR_FLAG
C1
10uF
C2
100nF
R1
910k
C3
100nF
R2
10k
R3
10k
R4
5.1k
R5
5.1k
R6
150k
VBUS_SENSE
R7
200k
TP1
VBUS_5V
TP2
USB0_DP
TP3
USB0_DM

Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication

STRUTHIO ARC-1

Sheet: /usb_power/
File: usb_power.kicad_sch

Title: ARC-1 Compact R8.5: USB-C sink (no host), VBUS protection, USB0 d

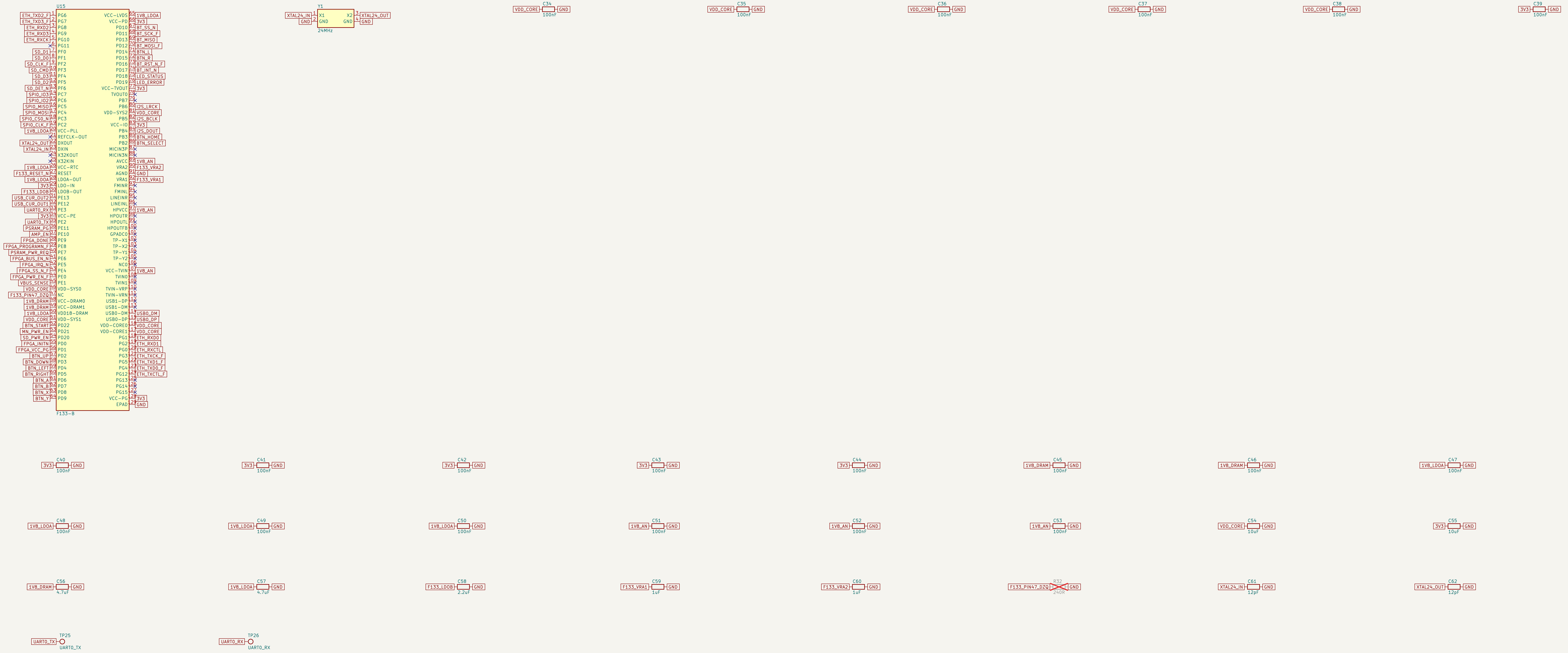
Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 2/10

Regulators, rail order, supervisors, staged−power links
Sheet generated from design.py; global labels carry every net.

U3 VBUS_5V 1 EN GND 2 GND SW_3V3 3 SW VIN 4 VBUS_5V PG 5 PG_3V3_BUCK FB 6 FB_3V3 3V3 2A
U4 PG_3V3 1 RESET GND 2 GND SUP33_MR_N 3 MR CT 4 SUP33_CT SENSE 5 3V3 VDD 6 3V3 TPS3808G33
U5 PG_3V3 1 EN GND 2 GND SW_CORE 3 SW VIN 4 VBUS_5V FB 5 FB_CORE VDD_CORE 0.95V
U6 PG_3V3 1 EN GND 2 GND SW_DRAM 3 SW VIN 4 VBUS_5V FB 5 FB_DRAM 1V8_DRAM
U7 3V3 1 IN GND 2 GND 3V3 3 EN NC 4 OUT 5 1V8_AN 1V8_AN
U8 F133_RESET_N 1 RESET GND 2 GND RST_BTN_N 3 MR CT 4 SUPC_CT SENSE 5 VDD_CORE VDD 6 3V3 TPS3808G09
U9 FPGA_PWR_EN 1 EN GND 2 GND SW_1V1 3 SW VIN 4 VBUS_5V PG 5 FPGA_VCC_PG FB 6 FB_1V1 1V1_FPGA
U10 3V3 1 VIN GND 2 GND FPGA_PWR_EN 3 ON CT 4 SW_FPGA_CT QOD 5 3V3_FPGA VOUT 6 3V3_FPGA 3V3_FPGA switch
U30 3V3_FPGA 1 IN GND 2 GND 3V3_FPGA 3 EN NC 4 OUT 5 2V5_AUX_REG 2V5_FPGA_AUX
U11 3V3 1 VIN GND 2 GND MN_PWR_EN 3 ON CT 4 SW_MN_CT QOD 5 3V3_MN VOUT 6 3V3_MN 3V3_MN switch
U12 3V3_FPGA 1 VIN GND 2 GND PSRAM_EN 3 ON CT 4 SW_PS_CT QOD 5 PSRAM_VDD VOUT 6 PSRAM_VDD PSRAM_VDD switch
U13 PSRAM_PWR_REQ 1 A DONE_DLY 2 B GND 3 GND Y 4 PSRAM_EN VCC 5 3V3 AND
U14 PSRAM_PG 1 RESET GND 2 GND PSRAM_SUP_MR_N 3 MR CT 4 PSRAM_SUP_CT SENSE 5 PSRAM_VDD VDD 6 3V3_FPGA TPS3808G30
J11 3V3 1 1 FPGA_PWR_EN 2 2 3 3 GND FPGA power override
SW1 RST_BTN_N 1 1 2 2 GND RESET
C4 VBUS_5V GND 10uF
L1 SW_3V3 3V3_REG 2.2uH
R8 3V3_REG FB_3V3 453k
R9 FB_3V3 GND 100k
C5 3V3_REG GND 22uF
C6 3V3_REG GND 22uF
R10 PG_3V3_BUCK 3V3_REG 100k
#FLG3 PWR_FLAG 3V3_REG
R11 3V3_REG 3V3 0R
#FLG4 PWR_FLAG 3V3
TP4 3V3_REG 3V3_REG
TP5 3V3 3V3
TP6 PG_3V3_BUCK PG_3V3_BUCK
C7 SUP33_CT GND 2.2nF
R12 SUP33_MR_N 3V3 100k
R13 PG_3V3 3V3 10k
C8 3V3 GND 100nF
C9 VBUS_5V GND 10uF
L2 SW_CORE VDD_CORE_REG 2.2uH
R14 VDD_CORE_REG FB_CORE 57.6k
R15 FB_CORE GND 100k
C10 VDD_CORE_REG GND 22uF
C11 VDD_CORE_REG GND 22uF
#FLG5 PWR_FLAG VDD_CORE_REG
R16 VDD_CORE_REG VDD_CORE 0R
#FLG6 PWR_FLAG VDD_CORE
TP7 VDD_CORE VDD_CORE
C12 VBUS_5V GND 10uF
L3 SW_DRAM 1V8_DRAM 2.2uH
R17 1V8_DRAM FB_DRAM 200k
R18 FB_DRAM GND 100k
C13 1V8_DRAM GND 22uF
#FLG7 PWR_FLAG 1V8_DRAM
TP8 1V8_DRAM 1V8_DRAM
C14 3V3 GND 1uF
C15 1V8_AN GND 1uF
TP9 1V8_AN 1V8_AN
C16 SUPC_CT GND 15nF
R19 RST_BTN_N 3V3 100k
R20 F133_RESET_N 1V8_LDOA 10k
TP10 F133_RESET_N F133_RESET_N
TP11 PG_3V3 PG_3V3
C17 VBUS_5V GND 10uF
L4 SW_1V1 1V1_FPGA_REG 2.2uH
R21 1V1_FPGA_REG FB_1V1 82.5k
R22 FB_1V1 GND 100k
C18 1V1_FPGA_REG GND 22uF
C19 1V1_FPGA_REG GND 22uF
#FLG8 PWR_FLAG 1V1_FPGA_REG
R23 1V1_FPGA_REG 1V1_FPGA 0R
#FLG9 PWR_FLAG 1V1_FPGA
R24 FPGA_PWR_EN_F FPGA_PWR_EN 1k
R25 FPGA_PWR_EN GND 100k
R26 FPGA_VCC_PG 3V3 100k
TP12 1V1_FPGA 1V1_FPGA
TP13 FPGA_VCC_PG FPGA_VCC_PG
TP14 FPGA_PWR_EN FPGA_PWR_EN
C20 SW_FPGA_CT GND 1nF
C21 3V3 GND 1uF
C22 3V3_FPGA GND 10uF
TP15 3V3_FPGA 3V3_FPGA
C23 3V3_FPGA GND 1uF
C24 2V5_AUX_REG GND 1uF
FB2 2V5_AUX_REG 2V5_FPGA_AUX ferrite 120R@100MHz
#FLG10 PWR_FLAG 2V5_FPGA_AUX
TP16 2V5_FPGA_AUX 2V5_FPGA_AUX
C25 SW_MN_CT GND 1nF
C26 3V3 GND 1uF
C27 3V3_MN GND 22uF
R27 MN_PWR_EN 3V3 100k
TP17 3V3_MN 3V3_MN
C28 SW_PS_CT GND 1nF
C29 3V3_FPGA GND 1uF
C30 PSRAM_VDD GND 4.7uF
C31 3V3 GND 100nF
R28 PSRAM_PWR_REQ GND 100k
R29 PSRAM_EN GND 100k
C32 PSRAM_SUP_CT GND 1nF
R30 PSRAM_SUP_MR_N 3V3_FPGA 100k
R31 PSRAM_PG 3V3_FPGA 10k
C33 3V3_FPGA GND 100nF
TP18 PSRAM_VDD PSRAM_VDD
TP19 PSRAM_PG PSRAM_PG
TP20 PSRAM_EN PSRAM_EN
TP21 GND GND
TP22 GND GND
TP23 GND GND
TP24 GND GND
#FLG11 PWR_FLAG GND
FID1 fiducial
FID2 fiducial
FID3 fiducial
H1 GND M2.5
H2 GND M2.5
H3 GND M2.5
H4 GND M2.5

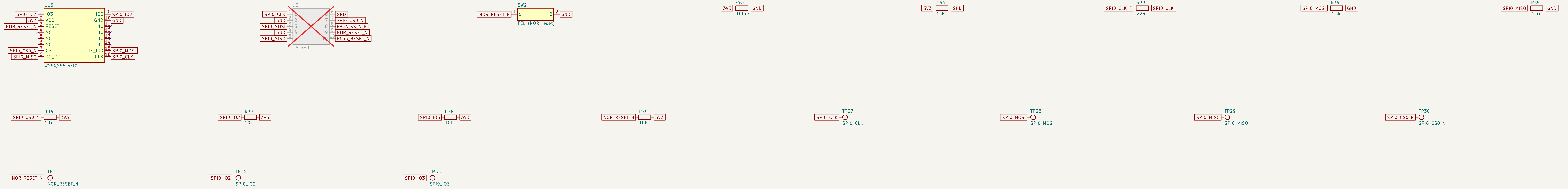
Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC−1
Sheet: /power/
File: power.kicad_sch
Title: ARC−1 Compact R8.5: Regulators, rail order, supervisors, staged−powe
Size: User
Date: 2026−10−15
Rev: R85−draft
KiCad E.D.A. 9.0.9
Id: 3/10

F133-B SoC: supplies, clock, reset, pin assignment
Sheet generated from design.py; global labels carry every net.


U15
ETH_TXD2_F 1 PG6
ETH_TXD3_F 2 PG7
ETH_RXD2 3 PG8
ETH_RXD3 4 PG9
ETH_RXCK 5 PG10
6 PG11
SD_D1 7 PF0
SD_D0 8 PF1
SD_CLK_F 9 PF2
SD_CMD 10 PF3
SD_D3 11 PF4
SD_D2 12 PF5
SD_DET_N 13 PF6
SPI0_IO3 14 PC7
SPI0_IO2 15 PC6
SPI0_MISO 16 PC5
SPI0_MOSI 17 PC4
SPI0_CS0_N 18 PC3
SPI0_CLK_F 19 PC2
1V8_LDOA 20 VCC-PLL
21 REFCLK-OUT
XTAL24_OUT 22 DXOUT
XTAL24_IN 23 DXIN
24 X32KOUT
25 X32KIN
1V8_LDOA 26 VCC-RTC
F133_RESET_N 27 RESET
1V8_LDOA 28 LDOA-OUT
3V3 29 LDO-IN
F133_LDOB 30 LDOB-OUT
USB_CUR_OUT2 31 PE13
USB_CUR_OUT1 32 PE12
UART0_RX 33 PE3
3V3 34 VCC-PE
UART0_TX 35 PE2
PSRAM_PG 36 PE11
AMP_EN 37 PE10
FPGA_DONE 38 PE9
FPGA_PROGRAMN_F 39 PE8
PSRAM_PWR_REQ 40 PE7
FPGA_BUS_EN_N 41 PE6
FPGA_IRQ_N 42 PE5
FPGA_SS_N_F 43 PE4
FPGA_PWR_EN_F 44 PE0
VBUS_SENSE 45 PE1
VDD_CORE 46 VDD-SYS0
F133_PIN47_DZQ 47 NC
1V8_DRAM 48 VCC-DRAM0
1V8_DRAM 49 VCC-DRAM1
1V8_LDOA 50 VDD18-DRAM
VDD_CORE 51 VDD-SYS1
BTN_START 52 PD22
MN_PWR_EN 53 PD21
SD_PWR_EN 54 PD20
FPGA_INITN 55 PD0
FPGA_VCC_PG 56 PD1
BTN_UP 57 PD2
BTN_DOWN 58 PD3
BTN_LEFT 59 PD4
BTN_RIGHT 60 PD5
BTN_A 61 PD6
BTN_B 62 PD7
BTN_X 63 PD8
BTN_Y 64 PD9
VCC-LVDS 65 1V8_LDOA
VCC-PD 66 3V3
PD10 67 BT_SS_N
PD11 68 BT_SCK_F
PD13 69 BT_MISO
PD12 70 BT_MOSI_F
PD14 71 BTN_L
PD15 72 BTN_R
PD16 73 BT_RST_N_F
PD17 74 BT_INT_N
PD18 75 LED_STATUS
PD19 76 LED_ERROR
VCC-TVOUT 77 3V3
TVOUT0 78
PB7 79
PB6 80 I2S_LRCK
VDD-SYS2 81 VDD_CORE
PB5 82 I2S_BCLK
VCC-IO 83 3V3
PB4 84 I2S_DOUT
PB3 85 BTN_HOME
PB2 86 BTN_SELECT
MICIN3P 87
MICIN3N 88
AVCC 89 1V8_AN
VRA2 90 F133_VRA2
AGND 91 GND
VRA1 92 F133_VRA1
FMINR 93
FMINL 94
LINEINR 95
LINEINL 96
HPVCC 97 1V8_AN
HPOUTR 98
HPOUTL 99
HPOUTFB 100
GPADC0 101
TP-X1 102
TP-X2 103
TP-Y1 104
TP-Y2 105
NC0 106
VCC-TVIN 107 1V8_AN
TVIN0 108
TVIN1 109
TVIN-VRP 110
TVIN-VRN 111
USB1-DP 112
USB1-DM 113
USB0-DM 114 USB0_DM
USB0-DP 115 USB0_DP
VDD-CORE0 116 VDD_CORE
VDD-CORE1 117 VDD_CORE
PG1 118 ETH_RXD0
PG2 119 ETH_RXD1
PG0 120 ETH_RXCTL
PG3 121 ETH_TXCK_F
PG5 122 ETH_TXD1_F
PG4 123 ETH_TXD0_F
PG12 124 ETH_TXCTL_F
PG13 125
PG14 126
PG15 127
VCC-PG 128 3V3
EPAD 129 GND
F133-B
Y1
XTAL24_IN 1 X1
GND 2 GND
X2 3 XTAL24_OUT
GND 4 GND
24MHz
VDD_CORE C34 100nF GND
VDD_CORE C35 100nF GND
VDD_CORE C36 100nF GND
VDD_CORE C37 100nF GND
VDD_CORE C38 100nF GND
3V3 C39 100nF GND
3V3 C40 100nF GND
3V3 C41 100nF GND
3V3 C42 100nF GND
3V3 C43 100nF GND
3V3 C44 100nF GND
1V8_DRAM C45 100nF GND
1V8_DRAM C46 100nF GND
1V8_LDOA C47 100nF GND
1V8_LDOA C48 100nF GND
1V8_LDOA C49 100nF GND
1V8_LDOA C50 100nF GND
1V8_AN C51 100nF GND
1V8_AN C52 100nF GND
1V8_AN C53 100nF GND
VDD_CORE C54 10uF GND
3V3 C55 10uF GND
1V8_DRAM C56 4.7uF GND
1V8_LDOA C57 4.7uF GND
F133_LDOB C58 2.2uF GND
F133_VRA1 C59 1uF GND
F133_VRA2 C60 1uF GND
F133_PIN47_DZQ R32 240R GND
XTAL24_IN C61 12pF GND
XTAL24_OUT C62 12pF GND
UART0_TX TP25 UART0_TX
UART0_RX TP26 UART0_RX

Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /f133/
File: f133.kicad_sch
Title: ARC-1 Compact R8.5: F133-B SoC: supplies, clock, reset, pin assignm
Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 4/10

32 MiB SPI NOR boot (BOOT-SEL 00), straps, FEL/clip recovery
Sheet generated from design.py; global labels carry every net.


U16
SPI0_IO3
3V3
NOR_RESET_N
SPI0_CS0_N
SPI0_MISO
1
2
3
4
5
6
7
8
IO3
VCC
RESET
NC
NC
NC
CS
DO_IO1
IO2
GND
NC
NC
NC
NC
DI_IO0
CLK
9
10
11
12
13
14
15
16
SPI0_IO2
GND
SPI0_MOSI
SPI0_CLK
W25Q256JVFIQ
J2
SPI0_CLK
GND
SPI0_MOSI
GND
SPI0_MISO
1
2
3
4
5
6
7
8
9
10
GND
SPI0_CS0_N
FPGA_SS_N_F
NOR_RESET_N
F133_RESET_N
LA SPI0
SW2
NOR_RESET_N
1
2
GND
FEL (NOR reset)
C63
3V3
GND
100nF
C64
3V3
GND
1uF
R33
SPI0_CLK_F
SPI0_CLK
22R
R34
SPI0_MOSI
GND
3.3k
R35
SPI0_MISO
GND
3.3k
R36
SPI0_CS0_N
3V3
10k
R37
SPI0_IO2
3V3
10k
R38
SPI0_IO3
3V3
10k
R39
NOR_RESET_N
3V3
10k
TP27
SPI0_CLK
SPI0_CLK
TP28
SPI0_MOSI
SPI0_MOSI
TP29
SPI0_MISO
SPI0_MISO
TP30
SPI0_CS0_N
SPI0_CS0_N
TP31
NOR_RESET_N
NOR_RESET_N
TP32
SPI0_IO2
SPI0_IO2
TP33
SPI0_IO3
SPI0_IO3

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SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /boot_nor/
File: boot_nor.kicad_sch
Title: ARC-1 Compact R8.5: 32 MiB SPI NOR boot (BOOT-SEL 00), straps, F
Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 5/10

MN820 (BT820B + 128 MiB DDR3L) module socket, host SPI1, LVDS to R09 panel
Sheet generated from design.py; global labels carry every net.

J3
1 GND — GND
3 GND — GND
5 NC
7 NC
9 NC
11 NC
13 VCC3V3 — 3V3_MN
15 VCC3V3 — 3V3_MN
17 VCC3V3 — 3V3_MN
19 VCC3V3 — 3V3_MN
21 SD_CD
23 SD_D0
25 SD_D1
27 SD_D2
29 SD_D3
31 SD_CLK
33 SD_CMD
35 SPIM_MOSI
37 SPIM_MISO
39 SPIM_IO3
41 SPIM_IO2
43 SPIM_SS_N
45 SPIM_SCLK
47 GND — GND
49 GND — GND
51 GND — GND
53 GND — GND
55 GND — GND
57 LVDS_TX0_D0_N
59 LVDS_TX0_D0_P
61 LVDS_TX0_D1_N
63 LVDS_TX0_D1_P
65 LVDS_TX0_D2_N
67 LVDS_TX0_D2_P
69 LVDS_TX0_CLK_N
71 LVDS_TX0_CLK_P
73 LVDS_TX0_D3_N
75 LVDS_TX0_D3_P
77 GND — GND
79 LVDS_TX1_D0_N
81 LVDS_TX1_D0_P
83 LVDS_TX1_D1_N
85 LVDS_TX1_D1_P
87 LVDS_TX1_D2_N
89 LVDS_TX1_D2_P
91 LVDS_TX1_CLK_N
93 LVDS_TX1_CLK_P
95 LVDS_TX1_D3_N
97 LVDS_TX1_D3_P
99 GND — GND
2 GND — GND
4 LVDS_RX0_D0_N
6 LVDS_RX0_D0_P
8 LVDS_RX0_D1_N
10 LVDS_RX0_D1_P
12 LVDS_RX0_D2_N
14 LVDS_RX0_D2_P
16 LVDS_RX0_CLK_N
18 LVDS_RX0_CLK_P
20 LVDS_RX0_D3_N
22 LVDS_RX0_D3_P
24 GND — GND
26 LVDS_RX1_D0_N
28 LVDS_RX1_D0_P
30 LVDS_RX1_D1_N
32 LVDS_RX1_D1_P
34 LVDS_RX1_D2_N
36 LVDS_RX1_D2_P
38 LVDS_RX1_CLK_N
40 LVDS_RX1_CLK_P
42 LVDS_RX1_D3_N
44 LVDS_RX1_D3_P
46 GND — GND
48 AUDIO_R
50 GND — GND
52 AUDIO_L
54 GND — GND
56 I2S_LRCLK
58 I2S_BCLK
60 I2S_SDA
62 DISP
64 BKLIT_PWM
66 GND — GND
68 GPIO3
70 GPIO2
72 GPIO4
74 GPIO6
76 GPIO5
78 GPIO7
80 GPIO8
82 GND — GND
84 SPIS_MISO — BT_MISO_M
86 SPIS_SCLK — BT_SCK
88 SPIS_SS_N — BT_SS_N_M
90 SPIS_MOSI — BT_MOSI
92 SPIS_IO3 — BT_IO3_M
94 SPIS_IO2 — BT_IO2_M
96 INT_N — BT_INT_N
98 RESET_N — BT_RST_N
100 GND — GND
DF40C-100DS-0.4V(51)
J4
BT_SCK 1
GND 2
BT_MOSI 3
GND 4
BT_MISO 5
6 GND
7 BT_SS_N
8 BT_INT_N
9 BT_RST_N
10 GND
LA SPI1
U17
BT_SS_N 1 $\overline{OE}$
BT_MISO_M 2 A
GND 3 GND
Y 4 BT_MISO
VCC 5 3V3
MISO buffer
BT_SCK_F R40 33R BT_SCK
BT_MOSI_F R41 33R BT_MOSI
BT_SS_N R42 33R BT_SS_N_M
BT_RST_N_F R43 33R BT_RST_N
BT_SS_N_M R44 10k 3V3_MN
BT_INT_N R45 10k 3V3_MN
BT_RST_N R46 10k GND
BT_MISO_M R47 10k GND
BT_IO2_M R48 10k GND
BT_IO3_M R49 10k GND
BT_MISO R50 47k GND
3V3 C65 100nF GND
3V3_MN C66 100nF GND
3V3_MN C67 100nF GND
3V3_MN C68 10uF GND
H5 MN820 hole
H6 MN820 hole
BT_SCK TP34 BT_SCK
BT_MISO TP35 BT_MISO
BT_INT_N TP36 BT_INT_N
BT_RST_N TP37 BT_RST_N

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SELECTED_UNVERIFIED engineering capture — NOT released for fabrication

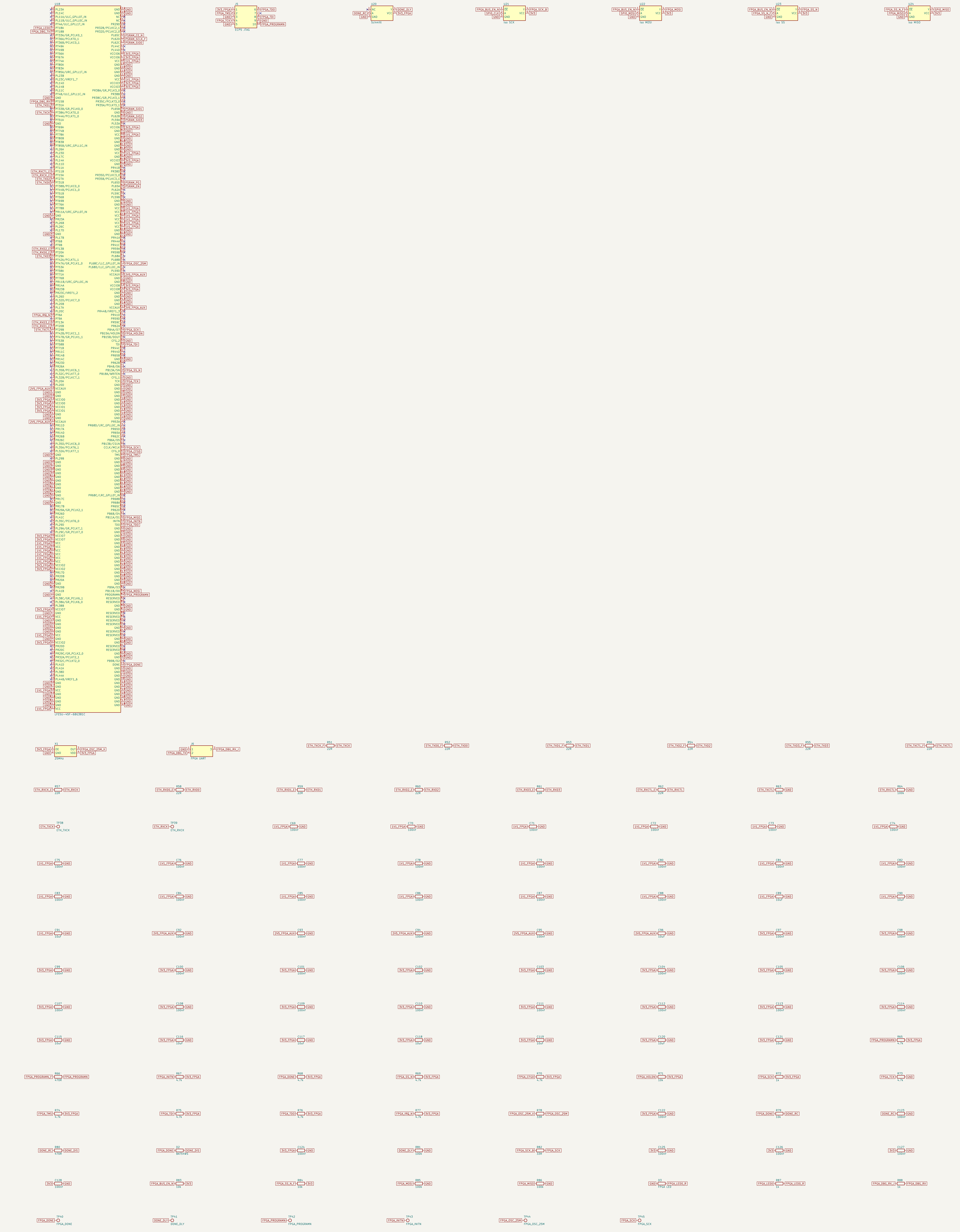
STRUTHIO ARC-1

Sheet: /mn820/
File: mn820.kicad_sch

Title: ARC-1 Compact R8.5: MN820 (BT820B + 128 MiB DDR3L) module sock

Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 6/10

LFE5U-45F-6BG381C (optional accelerator): rails, sysCONFIG slave SPI, JTAG, SPI0 isolation
Sheet generated from design.py; global labels carry every net.


U18
LFE5U-45F-6BG381C
PL23A
PL14C
PL11A/ULC_GPLL0T_IN
PL11B/ULC_GPLL0C_IN
PT4A/ULC_GPLL1T_IN
PT18A
PT18B
PT33A/GR_PCLK0_1
PT36A/PCLKT0_1
PT36B/PCLKC0_1
PT49A
PR29D
PR32B/PCLKC2_1
PR32D/PCLKC2_0
PL65C
PL62D
PL62C
PL44C
NC
FPGA_LED0
FPGA_DBG_TX
PSRAM_CE_N
PSRAM_SCLK_F
PSRAM_SIO0
PL38D
PL44A
PL44B/VREF1_6
VCC
GND
1V1_FPGA
J5
ECP5 JTAG
3V3_FPGA
FPGA_TMS
FPGA_TCK
FPGA_TDO
FPGA_TDI
FPGA_PROGRAMN
U20
Schmitt
NC
A
Y
VCC
DONE_RC
DONE_DLY
U21
iso SCK
OE
FPGA_BUS_EN_N
SPI0_CLK
FPGA_SCK_B
3V3
U22
iso MOSI
SPI0_MOSI
FPGA_MOSI
U23
iso SS
FPGA_SS_N_F
FPGA_SS_N
U24
iso MISO
FPGA_MISO
SPI0_MISO
X1
25MHz
OE
OUT
VDD
FPGA_OSC_25M_X
J6
FPGA UART
FPGA_DBG_RX_J
R51 22R ETH_TXCK_F ETH_TXCK
R52 22R ETH_TXD0_F ETH_TXD0
R53 22R ETH_TXD1_F ETH_TXD1
R54 22R ETH_TXD2_F ETH_TXD2
R55 22R ETH_TXD3_F ETH_TXD3
R56 22R ETH_TXCTL_F ETH_TXCTL
R57 22R ETH_RXCK_E ETH_RXCK
R58 22R ETH_RXD0_E ETH_RXD0
R59 22R ETH_RXD1_E ETH_RXD1
R60 22R ETH_RXD2_E ETH_RXD2
R61 22R ETH_RXD3_E ETH_RXD3
R62 22R ETH_RXCTL_E ETH_RXCTL
R63 100k ETH_TXCTL GND
R64 100k ETH_RXCTL GND
TP38 ETH_TXCK
TP39 ETH_RXCK
C69 100nF 1V1_FPGA GND
C70 100nF 1V1_FPGA GND
C71 100nF 1V1_FPGA GND
C72 100nF 1V1_FPGA GND
C73 100nF 1V1_FPGA GND
C74 100nF 1V1_FPGA GND
C75 100nF 1V1_FPGA GND
C76 100nF 1V1_FPGA GND
C77 100nF 1V1_FPGA GND
C78 100nF 1V1_FPGA GND
C79 100nF 1V1_FPGA GND
C80 100nF 1V1_FPGA GND
C81 100nF 1V1_FPGA GND
C82 100nF 1V1_FPGA GND
C83 100nF 1V1_FPGA GND
C84 100nF 1V1_FPGA GND
C85 100nF 1V1_FPGA GND
C86 100nF 1V1_FPGA GND
C87 100nF 1V1_FPGA GND
C88 100nF 1V1_FPGA GND
C89 10uF 1V1_FPGA GND
C90 10uF 1V1_FPGA GND
C91 10uF 1V1_FPGA GND
C92 100nF 2V5_FPGA_AUX GND
C93 100nF 2V5_FPGA_AUX GND
C94 100nF 2V5_FPGA_AUX GND
C95 100nF 2V5_FPGA_AUX GND
C96 10uF 2V5_FPGA_AUX GND
C97 100nF 3V3_FPGA GND
C98 100nF 3V3_FPGA GND
C99 100nF 3V3_FPGA GND
C100 100nF 3V3_FPGA GND
C101 100nF 3V3_FPGA GND
C102 100nF 3V3_FPGA GND
C103 100nF 3V3_FPGA GND
C104 100nF 3V3_FPGA GND
C105 100nF 3V3_FPGA GND
C106 100nF 3V3_FPGA GND
C107 100nF 3V3_FPGA GND
C108 100nF 3V3_FPGA GND
C109 100nF 3V3_FPGA GND
C110 100nF 3V3_FPGA GND
C111 100nF 3V3_FPGA GND
C112 100nF 3V3_FPGA GND
C113 100nF 3V3_FPGA GND
C114 100nF 3V3_FPGA GND
C115 10uF 3V3_FPGA GND
C116 10uF 3V3_FPGA GND
C117 10uF 3V3_FPGA GND
C118 10uF 3V3_FPGA GND
C119 10uF 3V3_FPGA GND
C120 10uF 3V3_FPGA GND
C121 10uF 3V3_FPGA GND
R65 4.7k FPGA_PROGRAMN 3V3_FPGA
R66 470R FPGA_PROGRAMN_F FPGA_PROGRAMN
R67 4.7k FPGA_INITN 3V3_FPGA
R68 4.7k FPGA_DONE 3V3_FPGA
R69 4.7k FPGA_SS_N 3V3_FPGA
R70 4.7k FPGA_CFG0 3V3_FPGA
R71 10k FPGA_HOLDN 3V3_FPGA
R72 1k FPGA_SCK 3V3_FPGA
R73 4.7k FPGA_TCK GND
R74 4.7k FPGA_TMS 3V3_FPGA
R75 4.7k FPGA_TDI 3V3_FPGA
R76 4.7k FPGA_TDO 3V3_FPGA
R77 4.7k FPGA_IRQ_N 3V3_FPGA
R78 33R FPGA_OSC_25M_X FPGA_OSC_25M
C122 100nF 3V3_FPGA GND
R79 10k FPGA_DONE DONE_RC
C123 100nF DONE_RC GND
R80 470R DONE_RC DONE_DIS
D2 BAT54WS FPGA_DONE DONE_DIS
C124 100nF 3V3_FPGA GND
R81 100k DONE_DLY GND
R82 33R FPGA_SCK_B FPGA_SCK
C125 100nF 3V3 GND
C126 100nF 3V3 GND
C127 100nF 3V3 GND
C128 100nF 3V3 GND
R83 10k FPGA_BUS_EN_N 3V3
R84 10k FPGA_SS_N_F 3V3
R85 100k FPGA_MOSI GND
R86 100k FPGA_MISO GND
D3 FPGA LED GND FPGA_LED0_R
R87 1k FPGA_LED0 FPGA_LED0_R
R88 1k FPGA_DBG_RX_J FPGA_DBG_RX
TP40 FPGA_DONE
TP41 DONE_DLY
TP42 FPGA_PROGRAMN
TP43 FPGA_INITN
TP44 FPGA_OSC_25M
TP45 FPGA_SCK

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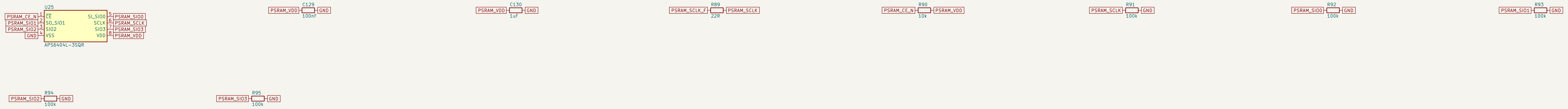
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /fpga/
File: fpga_ecp5.kicad_sch
Title: ARC-1 Compact R8.5: LFE5U-45F-6BG381C (optional accelerator): rai

Size: User	Date: 2026-10-15	**Rev: R85-draft**
KiCad E.D.A. 9.0.9		Id: 7/10

APS6404L 8 MiB PSRAM: switched VDD (DONE-gated), static pulls
Sheet generated from design.py; global labels carry every net.


U25
APS6404L-3SQR
1 CE
2 SO_SIO1
3 SIO2
4 VSS
5 SI_SIO0
6 SCLK
7 SIO3
8 VDD
PSRAM_CE_N
PSRAM_SIO1
PSRAM_SIO2
GND
PSRAM_SIO0
PSRAM_SCLK
PSRAM_SIO3
PSRAM_VDD
C129 100nF PSRAM_VDD GND
C130 1uF PSRAM_VDD GND
R89 22R PSRAM_SCLK_F PSRAM_SCLK
R90 10k PSRAM_CE_N PSRAM_VDD
R91 100k PSRAM_SCLK GND
R92 100k PSRAM_SIO0 GND
R93 100k PSRAM_SIO1 GND
R94 100k PSRAM_SIO2 GND
R95 100k PSRAM_SIO3 GND

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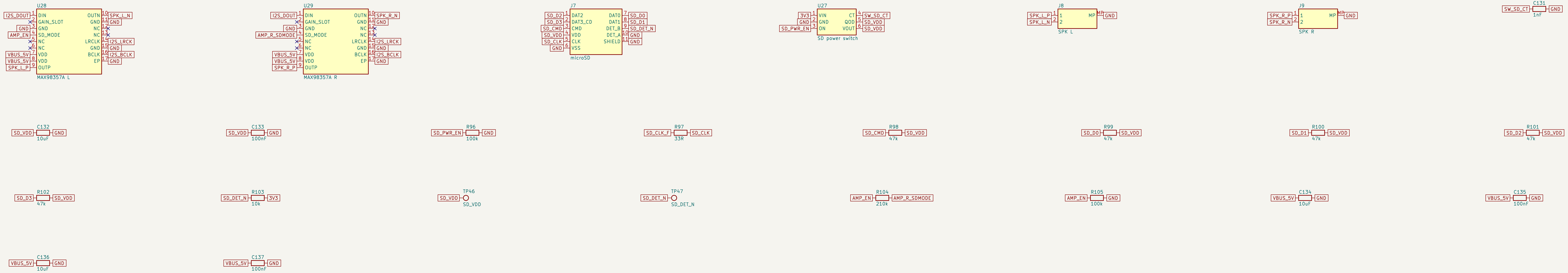
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /psram/
File: psram.kicad_sch
Title: ARC-1 Compact R8.5: APS6404L 8 MiB PSRAM: switched VDD (DONE-g
Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 8/10

microSD game media (not boot), 2x MAX98357A I2S speakers
Sheet generated from design.py; global labels carry every net.


U28
I2S_DOUT 1 DIN
2 GAIN_SLOT
GND 3 GND
AMP_EN 4 SD_MODE
5 NC
6 NC
VBUS_5V 7 VDD
VBUS_5V 8 VDD
SPK_L_P 9 OUTP
OUTN 10 SPK_L_N
GND 11 GND
NC 12
NC 13
LRCLK 14 I2S_LRCK
GND 15 GND
BCLK 16 I2S_BCLK
EP 17 GND
MAX98357A L
U29
I2S_DOUT 1 DIN
2 GAIN_SLOT
GND 3 GND
AMP_R_SDMODE 4 SD_MODE
5 NC
6 NC
VBUS_5V 7 VDD
VBUS_5V 8 VDD
SPK_R_P 9 OUTP
OUTN 10 SPK_R_N
GND 11 GND
NC 12
NC 13
LRCLK 14 I2S_LRCK
GND 15 GND
BCLK 16 I2S_BCLK
EP 17 GND
MAX98357A R
J7
SD_D2 1 DAT2
SD_D3 2 DAT3_CD
SD_CMD 3 CMD
SD_VDD 4 VDD
SD_CLK 5 CLK
GND 6 VSS
DAT0 7 SD_D0
DAT1 8 SD_D1
DET_B 9 SD_DET_N
DET_A 10 GND
SHIELD 11 GND
microSD
U27
3V3 1 VIN
GND 2 GND
SD_PWR_EN 3 ON
CT 4 SW_SD_CT
QOD 5 SD_VDD
VOUT 6 SD_VDD
SD power switch
J8
SPK_L_P 1 1
SPK_L_N 2 2
MP MP GND
SPK L
J9
SPK_R_P 1 1
SPK_R_N 2 2
MP MP GND
SPK R
SW_SD_CT C131 GND 1nF
SD_VDD C132 GND 10uF
SD_VDD C133 GND 100nF
SD_PWR_EN R96 GND 100k
SD_CLK_F R97 SD_CLK 33R
SD_CMD R98 SD_VDD 47k
SD_D0 R99 SD_VDD 47k
SD_D1 R100 SD_VDD 47k
SD_D2 R101 SD_VDD 47k
SD_D3 R102 SD_VDD 47k
SD_DET_N R103 3V3 10k
SD_VDD TP46 SD_VDD
SD_DET_N TP47 SD_DET_N
AMP_EN R104 AMP_R_SDMODE 210k
AMP_EN R105 GND 100k
VBUS_5V C134 GND 10uF
VBUS_5V C135 GND 100nF
VBUS_5V C136 GND 10uF
VBUS_5V C137 GND 100nF

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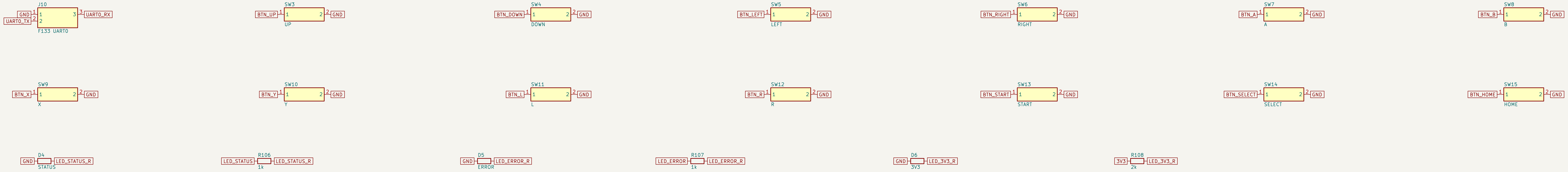
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /sd_audio/
File: sd_audio.kicad_sch
Title: ARC-1 Compact R8.5: microSD game media (not boot), 2x MAX98357A
Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 9/10

Game buttons, LEDs, debug UART
Sheet generated from design.py; global labels carry every net.


J10
GND
UART0_TX
UART0_RX
F133 UART0
SW3
BTN_UP
GND
UP
SW4
BTN_DOWN
GND
DOWN
SW5
BTN_LEFT
GND
LEFT
SW6
BTN_RIGHT
GND
RIGHT
SW7
BTN_A
GND
A
SW8
BTN_B
GND
B
SW9
BTN_X
GND
X
SW10
BTN_Y
GND
Y
SW11
BTN_L
GND
L
SW12
BTN_R
GND
R
SW13
BTN_START
GND
START
SW14
BTN_SELECT
GND
SELECT
SW15
BTN_HOME
GND
HOME
D4
GND
LED_STATUS_R
STATUS
R106
LED_STATUS
LED_STATUS_R
1k
D5
GND
LED_ERROR_R
ERROR
R107
LED_ERROR
LED_ERROR_R
1k
D6
GND
LED_3V3_R
3V3
R108
3V3
LED_3V3_R
2k

Generated by R85_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication

STRUTHIO ARC-1

Sheet: /controls/
File: controls.kicad_sch

Title: ARC-1 Compact R8.5: Game buttons, LEDs, debug UART

Size: User
Date: 2026-10-15
Rev: R85-draft
KiCad E.D.A. 9.0.9
Id: 10/10